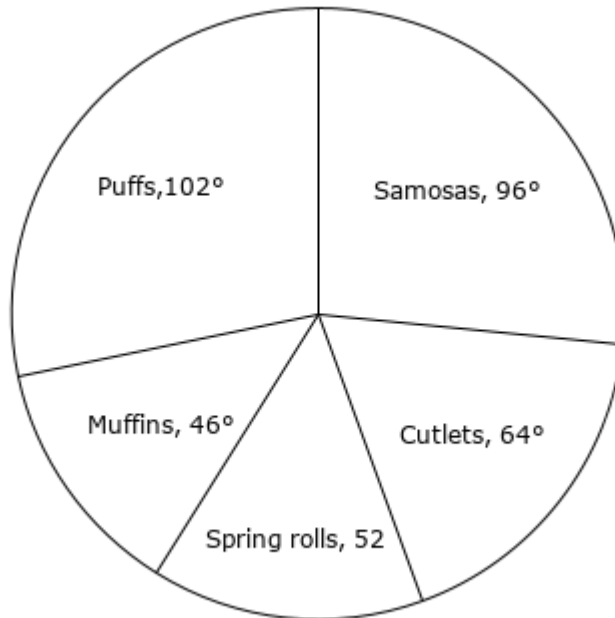


1. Questions

Study the following information carefully and answer the questions given below.

The pie chart below shows the degree distribution of the total number of five different snacks prepared for the function.



Note: The total number of snacks prepared for the function = 1800 @RankZenBanker

30% of the total number of muffins and 40% of the cutlets got wasted. Find the total number of muffins and cutlets that were not wasted.

- a. 349
- b. 325
- c. 343
- d. 353
- e. 359

2. Questions

The total number of cutlets prepared is what percentage more or less than the total number of samosas prepared for the function.

- a. 33.33% less
- b. 33.33% more
- c. 66.66% less
- d. 66.66% more
- e. 16.66% less

3. Questions

The ratio of the total number of Veg puffs prepared to the total number of Egg puffs prepared is 8:9. Find the total number of Egg puffs prepared for the function. (Only Veg and Egg puffs are prepared)

- a. 180
- b. 260
- c. 350
- d. 240
- e. 270

4. Questions

Another snack called vada was also prepared. The average of all six snacks is 340. Find the total number of vadas prepared for the function.

- a. 220
- b. 380
- c. 160
- d. 350
- e. 240

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5. Questions

Find the average of the total number of snacks prepared for the function, excluding spring rolls.

- a. 375
- b. 385
- c. 480
- d. 475
- e. 340

6. Questions

Study the following information carefully and answer the questions given below.

The data shows the total number of romance, thriller, and action movies released in April and May of 2026. The number of romance movies released in April is 25% more than the action movies released in May. The total number of movies released in May is 80% of the total number of movies released in April, which is 195. The total number of thriller movies released in April is 50% of the action movies released in April. The total number of action movies released in May is 48. Finally, the number of romance movies released in May is 70% more than the thriller movies released in May.

If 50% of the action movies released in May 2026 are hits and the others are flops, find the total number of action movies that are flops.

- a. 22
- b. 30
- c. 28
- d. 24
- e. 26

7. Questions

The total number of action movies released in March 2026 is 45% more than the total number of romance movies released in April 2026. Find the number of action movies released in March 2026.

- a. 75
- b. 87
- c. 89
- d. 93
- e. 72

8. Questions

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If in the next month, June, the number of romance movies increased by 12 and the number of thriller movies is the same with respect to May 2026, find the ratio of the number of thriller movies to romance movies in June.

- a. 2 : 1
- b. 3 : 1
- c. 1 : 3
- d. 2 : 3
- e. 1 : 2

9. Questions

Find the difference between the total number of romance movies released in April and May 2026, and the total number of action movies released in April 2026.

- a. 46
- b. 36
- c. 38
- d. 48

e. 28

10. Questions

The ratio of the total number of comedy movies released in April to the total number of thriller movies released in April is 9 : 5. Find the total number of comedy movies released in April 2026.

- a. 54
- b. 63
- c. 36
- d. 25
- e. 81

11. Questions

The ratio of the present ages of Q and P is 2:1. The ratio of P's age after 6 years to Q's age 4 years ago is 3:2. If the average age of P, Q, and R after 2 years will be 18 years, find the present age of S, which is half the present age of R.

- a. 24 years
- b. 20 years
- c. 15 years
- d. 18 years
- e. 30 years

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12. Questions

Mr. K invested Rs. x at a rate of 30% p.a., compounded annually for 2 years. He also invested Rs. $(x + 1500)$ at a simple interest rate of 7% p.a. for 3 years. If the total interest received from both investments is Rs. 450, find the value of x .

- a. 150
- b. 260
- c. 170
- d. 380
- e. 240

13. Questions

Pipe A and B can fill the tank together in 48 hours, while Pipe B and C can fill the tank together in 32 hours. Pipe C alone can fill the tank in 40 hours. In how many hours can Pipe A alone fill the tank?

- a. $68\frac{4}{7}$ hours

- b. $58\frac{4}{7}$ hours
- c. $64\frac{4}{7}$ hours
- d. $60\frac{4}{7}$ hours
- e. $48\frac{4}{7}$ hours

14. Questions

The area of a rectangle is 875 cm^2 , and its breadth is 35 cm. The base of a right-angled triangle is 4 cm less than the length of the rectangle, and its height is one-sixth of the perimeter of the rectangle. Find the area of the right-angled triangle.

- a. 380 cm^2
- b. 260 cm^2
- c. 340 cm^2
- d. 210 cm^2
- e. 420 cm^2

15. Questions

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The sum of two numbers, a and b, is 36, and the product of these two numbers is 308. The value of c is 50% more than the difference between a and b. If the average of a, b, c, and d is 16, find the value of d. (Note: $a > b$)

- a. 27
- b. 16
- c. 15
- d. 33
- e. 21

16. Questions

P invested Rs. $(x + 150)$ for 14 months, and after that, Q invested Rs. $(x - 250)$ for 10 months. After 2 years, the profit share of Q and P is 2 : 3. Find the value of P's initial investment.

- a. Rs. 5000
- b. Rs. 6000
- c. Rs. 8000
- d. Rs. 7000
- e. Rs. 4000

17. Questions

The total number of pens and pencils available in the shop is 216. The ratio of pens to pencils available is 5 : 4. The total number of items sold is 90, and the ratio of pencils to pens sold is 2 : 3. Find the ratio of the pens to pencils unsold in the shop.

- a. 10 : 11
- b. 11 : 10
- c. 11 : 9
- d. 9 : 11
- e. 11 : 12

18. Questions

The cost price of a bag is Rs. 1350. The shopkeeper marked it 60% above the cost price. After allowing two successive discounts of Rs. 126 and 50%, find the loss incurred by the shopkeeper.

- a. Rs. 243
- b. Rs. 433
- c. Rs. 333
- d. Rs. 144
- e. Rs. 323

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19. Questions

Vessel A contains 140 litres of a mixture of milk and water in the ratio of 2 : 3. Vessel B contains 260 litres of a mixture of milk and water in the ratio of 8 : 5. If they are all mixed together in vessel C, find the ratio of milk and water in vessel C.

- a. 22 : 17
- b. 27 : 23
- c. 23 : 27
- d. 13 : 22
- e. 17 : 22

20. Questions

Train A crosses a pole and a bridge of 600m in 32 seconds and 56 seconds respectively. The ratio of the speed of Train A to Train B is 5 : 4. If the time taken by Train B to cross a pole is 36 seconds, find the length of Train B.

- a. 680 m
- b. 960 m

- c. 720 m
- d. 540 m
- e. 820 m

21. Questions

What value should come in the place of (?) in the following questions.

$$(216 / 6^2) + (35 * 4) = (?) + 78 - 12$$

- a. 80
- b. 90
- c. 70
- d. 60
- e. 50

22. Questions

$$26\% \text{ of } 400 - 4^2 + (13 * 10) / 65 = (?) * 9$$

- a. 8
- b. 12
- c. 11
- d. 10
- e. 9

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23. Questions

$$(3(1/4) + 2(2/5)) * (?) = 200 - 99 + 5^3$$

- a. 30
- b. 70
- c. 40
- d. 50
- e. 60

24. Questions

$$2575 / 145) \div (103 / 29) + \sqrt[3]{1331} - 3^2 = (?)$$

- a. 4

- b. 6
- c. 5
- d. 7
- e. 8

25. Questions

$$(75^2 / 125) + 35\% \text{ of } 120 - (276 / 12) = (?)^3$$

- a. 3
- b. 4
- c. 5
- d. 6
- e. 7

26. Questions

What approximate value should come in the place of (?) in the following questions.

$$(98.12 / 13.97) + 22.88 - \sqrt[3]{510} = (?) / 7.11$$

- a. 264
- b. 176
- c. 132
- d. 198
- e. 154

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27. Questions

$$45.99\% \text{ of } 399.98 + (144.10 / 35.88) = (?) * 3.88$$

- a. 45
- b. 53
- c. 47
- d. 57
- e. 43

28. Questions

$$29.90\% \text{ of } 150.11 + (195.86 / 4.02) = (?)^2 - 6.12$$

- a. 12

- b. 10
- c. 14
- d. 6
- e. 8

29. Questions

$$(13.11 * 23.98) - (99.12 / 2.99) + \sqrt{120} = (?) * 5$$

- a. 44
- b. 64
- c. 58
- d. 68
- e. 54

30. Questions

$$(189.04 / 28.99) \div (8.91 / 87.03) = (?) - 25.11\% \text{ of } 35.88$$

- a. 72
- b. 74
- c. 76
- d. 64
- e. 62

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31. Questions

What value should come in the place of (?) in the following number series?

11, 12, 15, ?, 51, 132, 375

- a. 26
- b. 42
- c. 24
- d. 20
- e. 32

32. Questions

15, ?, 35, 65, 119, 205, 331

- a. 27

- b. 24
- c. 19
- d. 32
- e. 21

33. Questions**30030, 2310, 210, ?, 6, 2, 1**

- a. 28
- b. 65
- c. 50
- d. 30
- e. 42

34. Questions**22, ?, 49, 76, 112, 157, 211**

- a. 31
- b. 41
- c. 39
- d. 37
- e. 29

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35. Questions**2, 5, 11, 23, 47, ?, 191**

- a. 83
- b. 95
- c. 77
- d. 66
- e. 98

36. Questions**Find out the wrong number in the following number series.****121, 81, 54, 36, 24, 16**

- a. 81

- b. 121
- c. 54
- d. 36
- e. 16

37. Questions**12, 20, 36, 90, 270, 945, 3780**

- a. 270
- b. 20
- c. 12
- d. 36
- e. 90

38. Questions**30, 45, 32, 46, 28, 47, 27**

- a. 30
- b. 45
- c. 32
- d. 46
- e. 28

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39. Questions**4, 8, 14, 22, 32, 46, 58**

- a. 14
- b. 46
- c. 58
- d. 32
- e. 22

40. Questions**899, 778, 634, 465, 269, 44, 10**

- a. 10
- b. 44

- c. 269
- d. 465
- e. 778

41. Questions

The following question contains two equations as I and II. You have to solve both equations and determine the relationship between them and give an answer.

I: $3x^2 - 17x + 20 = 0$

II: $y^2 - 10y + 24 = 0$

- a. $x \geq y$
- b. $x > y$
- c. $x \leq y$
- d. $x = y$ or relationship cannot be established.
- e. $x < y$

42. Questions

I). $x^2 - 14x - 72 = 0$

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II). $y^2 + 20y + 96 = 0$

- a. $x \geq y$
- b. $x = y$ or relationship cannot be established.
- c. $x > y$
- d. $x < y$
- e. $x \leq y$

43. Questions

I). $5x + 3y = 26$

II). $6x + 5y = 20$

- a. $x = y$ or relationship cannot be established.
- b. $x \geq y$
- c. $x < y$
- d. $x \leq y$
- e. $x > y$

44. Questions

I). $x^2 + 11x + 30 = 0$

II). $y^2 - 7y - 18 = 0$

- a. $x \geq y$
- b. $x < y$
- c. $x = y$ or relationship cannot be established.
- d. $x > y$
- e. $x \leq y$

45. Questions

I). $(x + 8)^2 = 34x - 16$

II). $y^2 = 81$

- a. $x \geq y$
- b. $x > y$
- c. $x < y$
- d. $x \leq y$
- e. $x = y$ or relationship cannot be established.

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Explanations:

1. Questions

The total number of snacks prepared for the function = 1800

The total number of samosas prepared = $(96^\circ / 360^\circ) * 1800 = 480$

The total number of cutlets prepared = $(64^\circ / 360^\circ) * 1800 = 320$

The total number of spring rolls prepared = $(52^\circ / 360^\circ) * 1800 = 260$

The total number of muffins prepared = $(46^\circ / 360^\circ) * 1800 = 230$

The total number of puffs prepared = $(102^\circ / 360^\circ) * 1800 = 510$

Different types of snacks prepared	The total number of snacks prepared
Samosas	480
Cutlets	320
Spring rolls	260
Muffins	230
Puffs	510

Answer: D

The total number of muffins got wasted = 30% of 230

$$= (30 / 100) * 230 = 69$$

The total number of muffins that are not wasted = $230 - 69 = 161$

The total number of cutlets got wasted = 40% of 320

$$= (40 / 100) * 320 = 128$$

The total number of cutlets that are not wasted = $320 - 128 = 192$ @RankZenBanker

Required sum = $161 + 192 = 353$

Hence, the correct answer is option D

2. Questions

The total number of snacks prepared for the function = 1800

The total number of samosas prepared = $(96^\circ / 360^\circ) * 1800 = 480$

The total number of cutlets prepared = $(64^\circ / 360^\circ) * 1800 = 320$

The total number of spring rolls prepared = $(52^\circ / 360^\circ) * 1800 = 260$

The total number of muffins prepared = $(46^\circ / 360^\circ) * 1800 = 230$

The total number of puffs prepared = $(102^\circ / 360^\circ) * 1800 = 510$

Different types of snacks prepared	The total number of snacks prepared
Samosas	480
Cutlets	320
Spring rolls	260
Muffins	230
Puffs	510

Answer: A

Required percentage = $((480 - 320) / 480) * 100$

= $(160 / 480) * 100 = (1 / 3) * 100 = 33.33\%$ less

Hence, the correct answer is option A

3. Questions

The total number of snacks prepared for the function = 1800

The total number of samosas prepared = $(96^\circ / 360^\circ) * 1800 = 480$

The total number of cutlets prepared = $(64^\circ / 360^\circ) * 1800 = 320$

The total number of spring rolls prepared = $(52^\circ / 360^\circ) * 1800 = 260$

The total number of muffins prepared = $(46^\circ / 360^\circ) * 1800 = 230$

The total number of puffs prepared = $(102^\circ / 360^\circ) * 1800 = 510$

Different types of snacks prepared	The total number of snacks prepared
Samosas	480
Cutlets	320
Spring rolls	260
Muffins	230
Puffs	510

Answer: E

The total number of Veg puffs prepared : The total number of Egg puffs = 8 : 9

The total number of puffs prepared = 510

The total number of Egg puffs prepared = $510/17 * 9 = 270$

Hence, the correct answer is option E

4. Questions

The total number of snacks prepared for the function = 1800

The total number of samosas prepared = $(96^\circ / 360^\circ) * 1800 = 480$

The total number of cutlets prepared = $(64^\circ / 360^\circ) * 1800 = 320$

The total number of spring rolls prepared = $(52^\circ / 360^\circ) * 1800 = 260$

The total number of muffins prepared = $(46^\circ / 360^\circ) * 1800 = 230$

The total number of puffs prepared = $(102^\circ / 360^\circ) * 1800 = 510$

Different types of snacks prepared	The total number of snacks prepared
Samosas	480
Cutlets	320
Spring rolls	260
Muffins	230
Puffs	510

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Answer: E

Let the total number of vada prepared by x

Given that,

$$(480 + 320 + 260 + 230 + 510 + x) / 6 = 340$$

$$1800 + x = 2040$$

$$x = 240$$

Hence, the correct answer is option E

5. Questions

The total number of snacks prepared for the function = 1800

The total number of samosas prepared = $(96^\circ / 360^\circ) * 1800 = 480$

The total number of cutlets prepared = $(64^\circ / 360^\circ) * 1800 = 320$

The total number of spring rolls prepared = $(52^\circ / 360^\circ) * 1800 = 260$

The total number of muffins prepared = $(46^\circ / 360^\circ) * 1800 = 230$

The total number of puffs prepared = $(102^\circ / 360^\circ) * 1800 = 510$

Different types of snacks prepared	The total number of snacks prepared
Samosas	480
Cutlets	320
Spring rolls	260
Muffins	230
Puffs	510

Answer: B

Required average = $(480 + 320 + 230 + 510) / 4$

= $1540 / 4 = 385$

Hence, the correct answer is option B

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6. Questions

The total number of movies released in April = 195

The total number of movies released in May = 80% of the total number of movies released in April

= $(80 / 100) * 195 = 156$

The total number of romance movies released in April = 125% of the total number of action movies released in May

The total number of romance movies released in April : The total number of action movies released in May = 5 : 4

The total number of thriller movies released in April = 50% of the total number of action movies released in April

The total number of thriller movies released in April : The total number of action movies released in April = 1 : 2

The total number of action movies released in May = 48

The total number of romance movies released in May + The total number of thriller movies released in May = $156 - 48 = 108$

The total number of romance movies released in May = 170% of the total number of thriller movies released in May

The total number of romance movies released in May : The total number of thriller movies released in May = 17 : 10

The total number of romance movies released in May = $108/27 * 17 = 68$

The total number of thriller movies released in May = $108 - 68 = 40$

The total number of romance movies released in April = $(5 / 4) * 48 = 60$

The total number of thriller movies released in April + The total number of action movies released in April = $195 - 60 = 135$

The total number of thriller movies released in April = $135/3 * 1 = 45$

The total number of action movies released in April = $135 - 45 = 90$

Type of movies released	April 2026	May 2026
Romance movies	60	68
Thriller movies	45	40
Action movies	90	48

Answer: D

The total number of Action movies released in May 2026 = 48

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The total number of action movies that are hit = 50% of 48 = 24

The total number of action movies that are flop = $48 - 24 = 24$

Hence, the correct answer is option D

7. Questions

The total number of movies released in April = 195

The total number of movies released in May = 80% of the total number of movies released in April
 = $(80 / 100) * 195 = 156$

The total number of romance movies released in April = 125% of the total number of action movies released in May

The total number of romance movies released in April : The total number of action movies released in May = 5 : 4

The total number of thriller movies released in April = 50% of the total number of action movies released in April

The total number of thriller movies released in April : The total number of action movies released in April = 1 : 2

The total number of action movies released in May = 48

The total number of romance movies released in May + The total number of thriller movies released in May = $156 - 48 = 108$

The total number of romance movies released in May = 170% of the total number of thriller movies released in May

The total number of romance movies released in May : The total number of thriller movies released in May = $17 : 10$

The total number of romance movies released in May = $108/27 * 17 = 68$

The total number of thriller movies released in May = $108 - 68 = 40$

The total number of romance movies released in April = $(5 / 4) * 48 = 60$

The total number of thriller movies released in April + The total number of action movies released in April = $195 - 60 = 135$

The total number of thriller movies released in April = $135/3 * 1 = 45$

The total number of action movies released in April = $135 - 45 = 90$

Type of movies released	April 2026	May 2026
Romance movies	60	68
Thriller movies	45	40
Action movies	90	48

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Answer: B

The total number of action movies released in March 2026 = 145% of the total number of romance movies released in April 2026

The total number of action movies released in March 2026 = 145% of 60

= $(145 / 100) * 60 = 87$

Hence, the correct answer is option B

8. Questions

The total number of movies released in April = 195

The total number of movies released in May = 80% of the total number of movies released in April

= $(80 / 100) * 195 = 156$

The total number of romance movies released in April = 125% of the total number of action movies released in May

The total number of romance movies released in April : The total number of action movies released in May = $5 : 4$

The total number of thriller movies released in April = 50% of the total number of action movies released in April

The total number of thriller movies released in April : The total number of action movies released in April = 1 : 2

The total number of action movies released in May = 48

The total number of romance movies released in May + The total number of thriller movies released in May = 156 - 48 = 108

The total number of romance movies released in May = 170% of the total number of thriller movies released in May

The total number of romance movies released in May : The total number of thriller movies released in May = 17 : 10

The total number of romance movies released in May = $108/27 * 17 = 68$

The total number of thriller movies released in May = $108 - 68 = 40$

The total number of romance movies released in April = $(5 / 4) * 48 = 60$

The total number of thriller movies released in April + The total number of action movies released in April = $195 - 60 = 135$

The total number of thriller movies released in April = $135/3 * 1 = 45$

The total number of action movies released in April = $135 - 45 = 90$ **RankZenBanker**

Type of movies released	April 2026	May 2026
Romance movies	60	68
Thriller movies	45	40
Action movies	90	48

Answer: E

The total number of romance movies released in June 2026 = $68 + 12 = 80$

The total number of thriller movies released in June 2026 = 40

Required ratio = $40 : 80 = 1 : 2$

Hence, the correct answer is option E

9. Questions

The total number of movies released in April = 195

The total number of movies released in May = 80% of the total number of movies released in April
 = $(80 / 100) * 195 = 156$

The total number of romance movies released in April = 125% of the total number of action movies released in May

The total number of romance movies released in April : The total number of action movies released in May = 5 : 4

The total number of thriller movies released in April = 50% of the total number of action movies released in April

The total number of thriller movies released in April : The total number of action movies released in April = 1 : 2

The total number of action movies released in May = 48

The total number of romance movies released in May + The total number of thriller movies released in May = 156 - 48 = 108

The total number of romance movies released in May = 170% of the total number of thriller movies released in May

The total number of romance movies released in May : The total number of thriller movies released in May = 17 : 10

The total number of romance movies released in May = $108/27 * 17 = 68$

The total number of thriller movies released in May = $108 - 68 = 40$

The total number of romance movies released in April = $(5 / 4) * 48 = 60$

The total number of thriller movies released in April + The total number of action movies released in April = $195 - 60 = 135$

The total number of thriller movies released in April = $135/3 * 1 = 45$

The total number of action movies released in April = $135 - 45 = 90$

Type of movies released	April 2026	May 2026
Romance movies	60	68
Thriller movies	45	40
Action movies	90	48

Answer: C

Required difference = $(60 + 68) - 90 = 128 - 90 = 38$

Hence, the correct answer is option C

10. Questions

The total number of movies released in April = 195

The total number of movies released in May = 80% of the total number of movies released in April

$$= (80 / 100) * 195 = 156$$

The total number of romance movies released in April = 125% of the total number of action movies released in May

The total number of romance movies released in April : The total number of action movies released in May = 5 : 4

The total number of thriller movies released in April = 50% of the total number of action movies released in April

The total number of thriller movies released in April : The total number of action movies released in April = 1 : 2

The total number of action movies released in May = 48

The total number of romance movies released in May + The total number of thriller movies released in May = 156 - 48 = 108

The total number of romance movies released in May = 170% of the total number of thriller movies released in May

The total number of romance movies released in May : The total number of thriller movies released in May = 17 : 10

The total number of romance movies released in May = $108/27 * 17 = 68$

The total number of thriller movies released in May = $108 - 68 = 40$

The total number of romance movies released in April = $(5 / 4) * 48 = 60$

The total number of thriller movies released in April + The total number of action movies released in April = $195 - 60 = 135$

The total number of thriller movies released in April = $135/3 * 1 = 45$

The total number of action movies released in April = $135 - 45 = 90$

Type of movies released	April 2026	May 2026
Romance movies	60	68
Thriller movies	45	40
Action movies	90	48

Answer: E

The total number of comedy movies released in April = $(9 / 5) * 45 = 81$

Hence, the correct answer is option E

11. Questions

Answer: C

Let the present age of P, Q be x and $2x$ respectively.

According to the question,

$$(x + 6) / (2x - 4) = 3 / 2$$

$$2x + 12 = 6x - 12$$

$$12 + 12 = 6x - 2x$$

$$4x = 24$$

$$x = 6$$

The present age of P = $x = 6$ years

The present age of Q = $2x = 2 * 6 = 12$ years

Sum of the ages of P, Q and R after 2 years = $18 * 3 = 54$ years

Sum of the present ages of P, Q and R = $54 - 6 = 48$

Present age of R = $48 - (6 + 12) = 30$ years

Present age of S = $30/2 = 15$ years

Hence, the correct answer is option C

12. Questions

Answer: A

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Compound interest:

$$\text{Amount} = p * (1 + (r / 100))^n$$

$$\text{Amount} = x * (130 / 100)^2$$

$$\text{Amount} = x * (169 / 100)$$

$$\text{C.I} = (169 / 100) * x - x$$

$$\text{C.I} = x * (69 / 100)$$

Simple interest:

$$\text{S.I} = (p * n * r) / 100$$

$$\text{S.I} = ((x + 1500) * 3 * 7) / 100$$

$$\text{S.I} = (21x + 31500) / 100$$

Given that,

$$x * (69 / 100) + (21x + 31500) / 100 = 450$$

$$(90x + 31500) / 100 = 450$$

$$90x + 31500 = 45000$$

$$90x = 13500$$

$$x = 150$$

Hence, the correct answer is option A

13. Questions

Answer: A

Let the total capacity be 480 units (LCM of 48, 32, 40)

Efficiency of A and B = $480 / 48 = 10$ units

Efficiency of B and C = $480 / 32 = 15$ units

Efficiency of C = $480 / 40 = 12$ units

Efficiency of B = $15 - 12 = 3$ units

Efficiency of A = $10 - 3 = 7$ units

Pipe A alone fill the tank = $480 / 7 = 68(4/7)$ hours

Hence, the correct answer is option A

14. Questions

Answer: D

Let the length and breadth of the rectangle be l and b respectively

According to the question,

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$$(l * b) = 875$$

$$(l * 35) = 875$$

$$l = 875 / 35 = 25 \text{ cm}$$

$$\text{Perimeter of the rectangle} = 2 * (l + b)$$

$$= 2 * (25 + 35) = 120$$

$$\text{Base of the right-angled triangle} = \text{Length of the rectangle} - 4 = 25 - 4 = 21$$

$$\text{Height of the right-angled triangle} = (1 / 6) * \text{Perimeter of the rectangle}$$

$$= (1 / 6) * 120 = 20$$

$$\text{Area of the right-angled triangle} = (1 / 2) * b * h$$

$$= (1 / 2) * 21 * 20$$

$$= 210 \text{ cm}^2$$

Hence, the correct answer is option D

15. Questions

Answer: B

$$a + b = 36 \rightarrow (1)$$

$$a * b = 308$$

$$b = 308 / a \rightarrow (2)$$

Sub equation (2) in (1),

$$a + (308 / a) = 36$$

$$a^2 + 308 = 36a$$

$$a^2 - 36a + 308 = 0$$

$$a^2 - 22a - 14a + 308 = 0$$

$$a(a - 22) - 14(a - 22) = 0$$

$$(a - 14)(a - 22) = 0$$

$$a = 14, 22$$

Sub $a = 22$ in equation (2),

$$b = 308 / 22 = 14$$

$$a > b$$

So, $a = 22$ and $b = 14$

$$c = 150\% \text{ of } (a - b)$$

$$= (150 / 100) * (22 - 14)$$

$$= 12$$

Given that,

$$(a + b + c + d) / 4 = 16$$

$$(22 + 14 + 12 + d) / 4 = 16$$

$$48 + d = 64$$

$$d = 16$$

Hence, the correct answer is option B

16. Questions

Answer: B

According to the question,

$$((x + 150) * 14) / ((x - 250) * 10) = 3 / 2$$

$$(x + 150) * 14 / (x - 250) * 10 = 3 / 2$$

$$(x + 150) * 7 / (x - 250) * 5 = 3 / 2$$

$$(7x + 1050) / (5x - 1250) = 3 / 2$$

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$$14x + 2100 = 15x - 3750$$

$$15x - 14x = 2100 + 3750$$

$$x = 5850$$

The initial investment of P = $(x + 150)$

$$= 5850 + 150 = \text{Rs. } 6000$$

Hence, the correct answer is option B

17. Questions

Answer: B

The total number of pens and pencils available = 216

The total number of pens available : The total number of pencils available = 5 : 4

$$\text{The total number of pens available} = 216/9 * 5 = 120$$

$$\text{The total number of pencils available} = 216 - 120 = 96$$

$$\text{The total number of pens and pencils sold} = 90$$

$$\text{The total number of pens sold : The total number of pencils sold} = 3 : 2$$

$$\text{The total number of pens sold} = 90/5 * 3 = 54$$

$$\text{The total number of pencils sold} = 90 - 54 = 36$$

$$\text{The total number of pens unsold} = 120 - 54 = 66$$

$$\text{The total number of pencils unsold} = 96 - 36 = 60$$

$$\text{Required ratio} = 66 : 60 = 11 : 10$$

Hence, the correct answer is option B

18. Questions

Answer: C

The cost price of the bag = Rs. 1350

The marked price of the bag = 160% of the cost price of the bag

$$= (160 / 100) * 1350 = \text{Rs. } 2160$$

Marked price - Discount = Selling price

$$(2160 - 126) * (50 / 100) = \text{Selling price}$$

$$\text{Selling price} = \text{Rs. } 1017$$

Loss incurred = Cost price - Selling price

$$\text{Loss} = 1350 - 1017$$

$$= \text{Rs. } 333$$

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Hence, the correct answer is option C

19. Questions

Answer: B

Vessel A:

The total quantity of mixture = 140 litres

The total quantity of milk = $140/5 * 2 = 56$ litres

The total quantity of water = $140 - 56 = 84$ litres

Vessel B:

The total quantity of mixture = 260

The total quantity of milk = $260/13 * 8 = 160$ litres

The total quantity of water = $260 - 160 = 100$ litres

Vessel C:

The total quantity of milk = $56 + 160 = 216$ litres

The total quantity of water = $84 + 100 = 184$ liters

Required ratio = 216 : 184

= $54 : 46 = 27 : 23$

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Hence, the correct answer is option B

20. Questions

Answer: C

Let the length of Train A, and the speeds of Train A and Train B, be l , S_1 , and S_2 respectively

According to the question,

Distance = Speed * Time taken

$$l = S_1 * 32$$

$$l / 32 = S_1 \rightarrow (1)$$

$$(l + 600) = S_1 * 56$$

$$(l + 600) / 56 = S_1 \rightarrow (2)$$

Equate equation (1) and (2),

$$l / 32 = (l + 600) / 56$$

$$l / 8 = (l + 600) / 14$$

$$14l = 8l + 4800$$

$$6l = 4800$$

$$l = 800 \text{ m}$$

Substitute $l = 800 \text{ m}$ in equation (1)

$$800 = S_1 * 32$$

$$S_1 = 25 \text{ m/s}$$

The speed of the Train A = 25 m/s

The speed of the Train B = $S_2 = (4 / 5) * 25 = 20 \text{ m/s}$

$$\text{Distance} = 20 * 36 = 720 \text{ m}$$

The length of the Train B = 720 m

Hence, the correct answer is option C

21. Questions

Answer: A

$$(216 / 6^2) + (35 * 4) = (?) + 78 - 12$$

$$(216 / 36) + 140 = (?) + 66$$

$$6 + 140 = (?) + 66$$

$$146 - 66 = (?)$$

$$(?) = 80$$

Hence, the correct answer is option A

22. Questions

Answer: D

$$26\% \text{ of } 400 - 4^2 + (13 * 10) / 65 = (?) * 9$$

$$(26 / 100) * 400 - 16 + (130 / 65) = (?) * 9$$

$$104 - 16 + 2 = (?) * 9$$

$$90 / 9 = (?)$$

$$(?) = 10$$

Hence, the correct answer is option D

23. Questions

Answer: C

$$(3(1/4) + 2(2/5)) * (?) = 200 - 99 + 5^3$$

$$((13 / 4) + (12 / 5)) * (?) = 101 + 125$$

$$((65 + 48) / 20) * (?) = 226$$

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$$(113 / 20) * (?) = 226$$

$$(?) = (226 * 20) / 113$$

$$(?) = 40$$

Hence, the correct answer is option C

24. Questions

Answer: D

$$(2575 / 145) \div (103 / 29) + \sqrt[3]{1331} - 3^2 = (?)$$

$$(2575 / 145) * (29 / 103) + 11 - 9 = (?)$$

$$(515 / 29) * (29 / 103) + 2 = (?)$$

$$5 + 2 = (?)$$

$$(?) = 7$$

Hence, the correct answer is option D

25. Questions

Answer: B

$$(75^2 / 125) + 35\% \text{ of } 120 - (276 / 12) = (?)^3$$

$$(5625 / 125) + (35 / 100) * 120 - 23 = (?)^3$$

$$45 + 42 - 23 = (?)^3$$

$$64 = (?)^3$$

$$\sqrt[3]{64} = (?)$$

$$(?) = 4$$

Hence, the correct answer is option B

26. Questions

Answer: E

$$(98.12 / 13.97) + 22.88 - \sqrt[3]{510} = (?) / 7.11$$

$$(98 / 14) + 23 - \sqrt[3]{512} = (?) / 7$$

$$7 + 23 - 8 = (?) / 7$$

$$30 - 8 = (?) / 7$$

$$22 * 7 = (?)$$

$$(?) = 154$$

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Hence, the correct answer is option E

27. Questions

Answer: C

$$45.99\% \text{ of } 399.98 + (144.10 / 35.88) = (?) * 3.88$$

$$46\% \text{ of } 400 + (144 / 36) = (?) * 4$$

$$(46 / 100) * 400 + 4 = (?) * 4$$

$$184 + 4 = (?) * 4$$

$$188 / 4 = (?)$$

$$(?) = 47$$

Hence, the correct answer is option C

28. Questions

Answer: B

$$29.90\% \text{ of } 150.11 + (195.86 / 4.02) = (?)^2 - 6.12$$

$$30\% \text{ of } 150 + (196 / 4) = (?)^2 - 6$$

$$(30 / 100) * 150 + 49 + 6 = (?)^2$$

$$45 + 55 = (?)^2$$

$$\sqrt{100} = (?)$$

$$10 = (?)$$

Hence, the correct answer is option B

29. Questions

Answer: C

$$(13.11 * 23.98) - (99.12 / 2.99) + \sqrt{120} = (?) * 5$$

$$(13 * 24) - (99 / 3) + \sqrt{121} = (?) * 5$$

$$312 - 33 + 11 = (?) * 5$$

$$290 = (?) * 5$$

$$(?) = 58$$

Hence, the correct answer is option C

30. Questions

Answer: A

$$(189.04 / 28.99) \div (8.91 / 87.03) = (?) - 25.11\% \text{ of } 35.88$$

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$$(189 / 29) \div (9 / 87) = (?) - 25\% \text{ of } 36$$

$$(189 * 87) / (29 * 9) = (?) - (25 / 100) * 36$$

$$(16,443 / 261) = (?) - 9$$

$$63 + 9 = (?)$$

$$72 = (?)$$

Hence, the correct answer is option A

31. Questions

Answer: C

$$11 + 3^0 = 12$$

$$12 + 3^1 = 15$$

$$15 + 3^2 = \mathbf{24}$$

$$24 + 3^3 = 51$$

$$51 + 3^4 = 132$$

$$132 + 3^5 = 375$$

Hence, the correct answer is option C

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32. Questions

Answer: E

$$15 \quad \mathbf{21} \quad 35 \quad 65 \quad 119 \quad 205 \quad 331$$

$$6 \quad 14 \quad 30 \quad 54 \quad 86 \quad 126$$

$$8 \quad 16 \quad 24 \quad 32 \quad 40$$

Hence, the correct answer is option E

33. Questions

Answer: D

$$30030 / 13 = 2310$$

$$2310 / 11 = 210$$

$$210 / 7 = \mathbf{30}$$

$$30 / 5 = 6$$

$$6 / 3 = 2$$

$$2 / 2 = 1$$

Hence, the correct answer is option D

34. Questions

Answer: A

$$22 + (9 * 1) = 31$$

$$31 + (9 * 2) = 49$$

$$49 + (9 * 3) = 76$$

$$76 + (9 * 4) = 112$$

$$112 + (9 * 5) = 157$$

$$157 + (9 * 6) = 211$$

Hence, the correct answer is option A

35. Questions

Answer: B

$$(2 * 2) + 1 = 5$$

$$(5 * 2) + 1 = 11$$

$$(11 * 2) + 1 = 23$$

$$(23 * 2) + 1 = 47$$

$$(47 * 2) + 1 = 95$$

$$(95 * 2) + 1 = 191$$

Hence, the correct answer is option B

36. Questions

Answer: B

$$121.5 * (2 / 3) = 81$$

$$81 * (2 / 3) = 54$$

$$54 * (2 / 3) = 36$$

$$36 * (2 / 3) = 24$$

$$24 * (2 / 3) = 16$$

Hence, the correct answer is option B

37. Questions

Answer: B

$$12 * 1.5 = 18$$

$$18 * 2 = 36$$

$$36 * 2.5 = 90$$

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$$90 * 3 = 270$$

$$270 * 3.5 = 945$$

$$945 * 4 = 3780$$

Hence, the correct answer is option B

38. Questions

Answer: C

$$30 + 15 = 45$$

$$45 - 16 = \mathbf{29}$$

$$29 + 17 = 46$$

$$46 - 18 = 28$$

$$28 + 19 = 47$$

$$47 - 20 = 27$$

Hence, the correct answer is option C

39. Questions

Answer: B

$$(1 * 2) + 2 = 4$$

$$(2 * 3) + 2 = 8$$

$$(3 * 4) + 2 = 14$$

$$(4 * 5) + 2 = 22$$

$$(5 * 6) + 2 = 32$$

$$(6 * 7) + 2 = \mathbf{44}$$

$$(7 * 8) + 2 = 58$$

Another way:

$$4 \quad 8 \quad 14 \quad 22 \quad 32 \quad \mathbf{44} \quad 58$$

$$4 \quad 6 \quad 8 \quad 10 \quad 12 \quad 14$$

Hence, the correct answer is option B

40. Questions

Answer: A

$$899 - 11^2 = 778$$

$$778 - 12^2 = 634$$

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$$634 - 13^2 = 465$$

$$465 - 14^2 = 269$$

$$269 - 15^2 = 44$$

$$44 - 16^2 = -212$$

Hence, the correct answer is option A

41. Questions

Answer: C

I:

$$3x^2 - 12x - 5x + 20 = 0$$

$$3x(x - 4) - 5(x - 4) = 0$$

$$(3x - 5)(x - 4) = 0$$

$$x = 4, 1.66$$

II:

$$y^2 - 10y + 24 = 0$$

$$y^2 - 4y - 6y + 24 = 0$$

$$y(y - 4) - 6(y - 4) = 0$$

$$(y - 6)(y - 4) = 0$$

$$y = 4, 6$$

Hence, $x \leq y$

Hence, the correct answer is option C

42. Questions

Answer: C

I). $x^2 - 14x - 72 = 0$

$$x^2 + 4x - 18x - 72 = 0$$

$$x(x + 4) - 18(x + 4) = 0$$

$$(x - 18)(x + 4) = 0$$

$$x = 18, -4$$

II). $y^2 + 20y + 96 = 0$

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$$y^2 + 8y + 12y + 96 = 0$$

$$y(y + 8) + 12(y + 8) = 0$$

$$(y + 12)(y + 8) = 0$$

$$y = -12, -8$$

Hence, $x > y$

Hence, the correct answer is option C

43. Questions

Answer: E

I). $5x + 3y = 26$

$$5x = 26 - 3y$$

$$x = (26 - 3y) / 5$$

II). $6x + 5y = 20$

Sub $x = (26 - 3y) / 5$ in equation II

$$6 * ((26 - 3y) / 5) + 5y = 20$$

$$156 - 18y + 25y = 100$$

$$156 - 100 + 7y = 0$$

$$7y = -56$$

$$y = -8$$

Sub $y = -8$ in $x = (26 - 3y) / 5$

$$x = (26 - 3 * (-8)) / 5$$

$$= (26 + 24) / 5$$

$$= 50 / 5$$

$$x = 10$$

Hence, $x > y$

Hence, the correct answer is option E

44. Questions

Answer: B

I). $x^2 + 11x + 30 = 0$

$$x^2 + 5x + 6x + 30 = 0$$

$$x(x + 5) + 6(x + 5) = 0$$

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$$(x + 5)(x + 6) = 0$$

$$x = -5 \text{ or } -6$$

$$\text{II). } y^2 - 7y - 18 = 0$$

$$y^2 - 9y + 2y - 18 = 0$$

$$y(y - 9) + 2(y - 9) = 0$$

$$(y - 9)(y + 2) = 0$$

$$y = 9 \text{ or } -2$$

Hence, $x < y$

Hence, the correct answer is option B

45. Questions

Answer: E

$$\text{I). } (x + 8)^2 = 34x - 16$$

$$(x^2 + 16x + 64) = 34x - 16$$

$$x^2 + 16x - 34x + 64 + 16 = 0$$

$$x^2 - 18x + 80 = 0$$

$$x^2 - 8x - 10x + 80 = 0$$

$$x(x - 8) - 10(x - 8) = 0$$

$$(x - 10)(x - 8) = 0$$

$$x = 10, 8$$

$$\text{II). } y^2 = 81$$

$$y = \sqrt{81}$$

$$y = 9, -9$$

Hence, $x = y$ or relationship cannot be established.

Hence, the correct answer is option E

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